

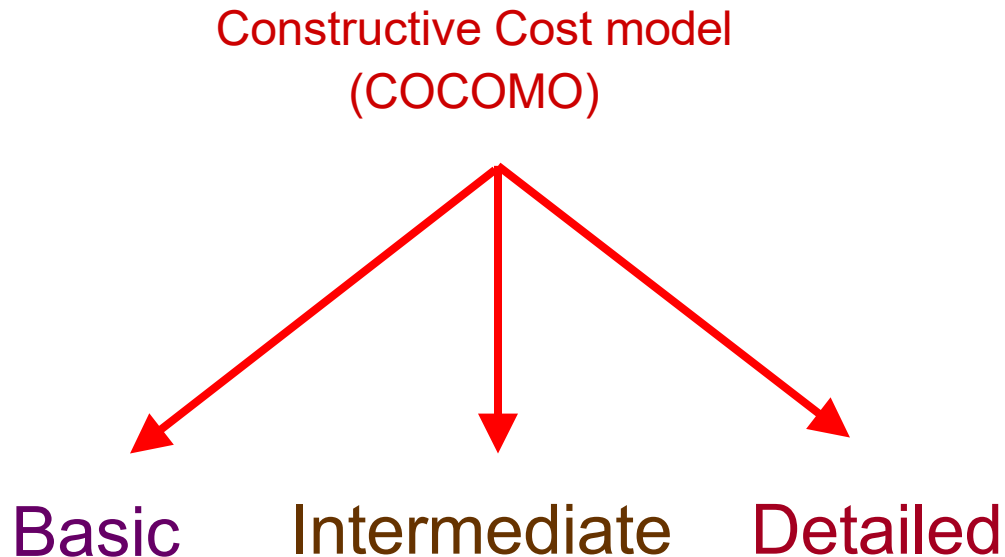
COCOMO Model

Constructive Cost Model

- **COCOMO Model**
- **Types of COCOMO Model**
- **COCOMO – II**

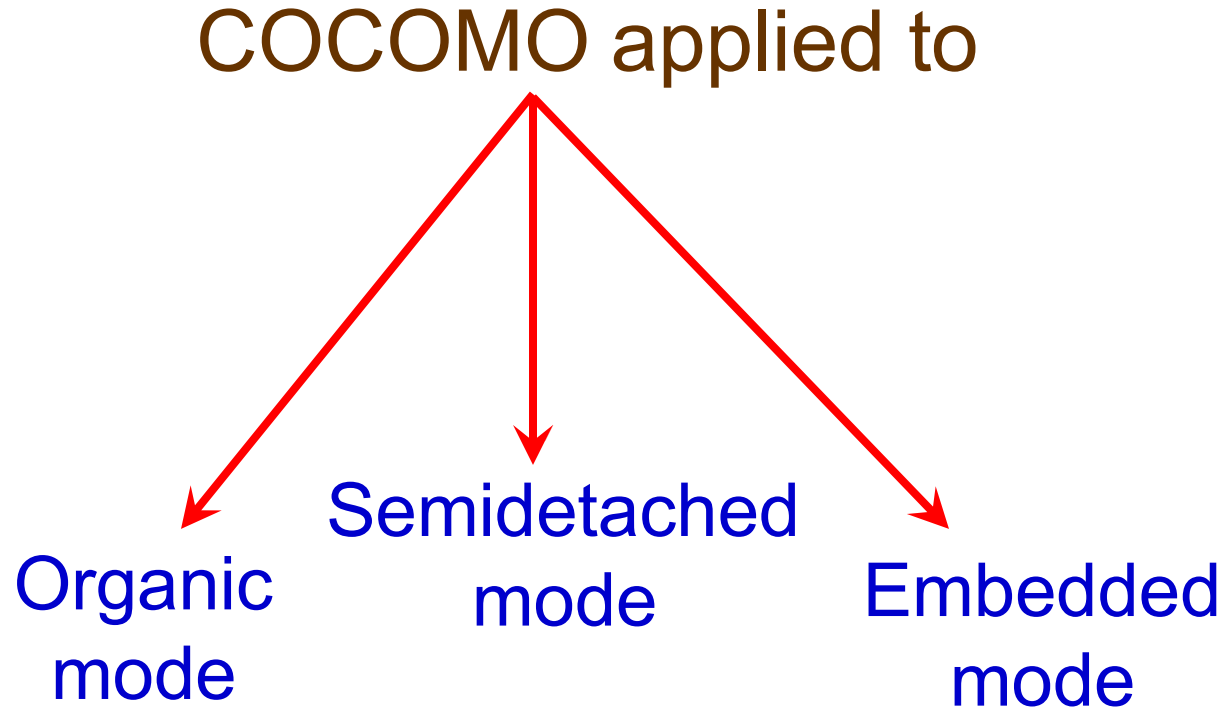
Software Project Planning

The Constructive Cost Model (COCOMO)



Model proposed by
B. W. Boehm's through his book
Software Engineering Economics in
1981

Software Project Planning



Software Project Planning

<i>Mode</i>	<i>Project size</i>	<i>Nature of Project</i>	<i>Innovation</i>	<i>Deadline of the project</i>	<i>Development Environment</i>
Organic	Typically 2-50 KLOC	Small size project, experienced developers in the familiar environment. For example, pay roll, inventory projects etc.	Little	Not tight	Familiar & In house
Semi detached	Typically 50-300 KLOC	Medium size project, Medium size team, Average previous experience on similar project. For example: Utility systems like compilers, database systems, editors etc.	Medium	Medium	Medium
Embedded	Typically over 300 KLOC	Large project, Real time systems, Complex interfaces, Very little previous experience. For example: ATMs, Air Traffic Control etc.	Significant	Tight	Complex Hardware/customer Interfaces required

Table 4: The comparison of three COCOMO modes

Software Project Planning

Basic Model

Basic COCOMO model takes the form

$$E = a_b(KLOC)^{b_b}$$

$$D = c_b(E)^{d_b}$$

where E is effort applied in Person-Months, and D is the development time in months. The coefficients a_b , b_b , c_b and d_b are given in table 4 (a).

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Software Project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Table 4(a): Basic COCOMO coefficients

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When effort and development time are known, the average staff size to complete the project may be calculated as:

$$\text{Average staff size} \quad (SS) = \frac{E}{D} \text{ Persons}$$

When project size is known, the productivity level may be calculated as:

$$\text{Productivity} \quad (P) = \frac{KLOC}{E} KLOC / PM$$

Software Project Planning

Example: 4.5

Suppose that a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three modes i.e., organic, semidetached and embedded.

Software Project Planning

Solution

The basic COCOMO equation take the form:

$$E = a_b (KLOC)^{b_b}$$

$$D = c_b (KLOC)^{d_b}$$

Estimated size of the project = 400 KLOC

(i) Organic mode

$$E = 2.4(400)^{1.05} = 1295.31 \text{ PM}$$

$$D = 2.5(1295.31)^{0.38} = 38.07 \text{ PM}$$

Software Project Planning

(ii) Semidetached mode

$$E = 3.0(400)^{1.12} = 2462.79 \text{ PM}$$

$$D = 2.5(2462.79)^{0.35} = 38.45 \text{ PM}$$

(iii) Embedded mode

$$E = 3.6(400)^{1.20} = 4772.81 \text{ PM}$$

$$D = 2.5(4772.8)^{0.32} = 38 \text{ PM}$$

Software Project Planning

Example: 4.6

A project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and productivity of the project.

Software Project Planning

Solution

The semi-detached mode is the most appropriate mode; keeping in view the size, schedule and experience of the development team.

Hence $E = 3.0(200)^{1.12} = 1133.12 \text{ PM}$

$$D = 2.5(1133.12)^{0.35} = 29.3 \text{ PM}$$

Average staff size $(SS) = \frac{E}{D} \text{ Persons}$

$$= \frac{1133.12}{29.3} = 38.67 \text{ Persons}$$

Software Project Planning

$$\text{Productivity} = \frac{KLOC}{E} = \frac{200}{1133.12} = 0.1765 \text{ KLOC} / \text{PM}$$

$$P = 176 \text{ LOC} / \text{PM}$$

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Intermediate Model

Cost drivers

i) Product Attributes

- Required s/w reliability
- Size of application database
- Complexity of the product

i) Hardware Attributes

- Run time performance constraints
- Memory constraints
- Virtual machine volatility
- Turnaround time

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(iii) Personal Attributes

- Analyst capability
- Programmer capability
- Application experience
- Virtual m/c experience
- Programming language experience

(iv) Project Attributes

- Modern programming practices
- Use of software tools
- Required development Schedule

Software Project Planning

Multipliers of different cost drivers

Cost Drivers	RATINGS					
	Very low	Low	Nominal	High	Very high	Extra high
Product Attributes						
RELY	0.75	0.88	1.00	1.15	1.40	--
DATA	--	0.94	1.00	1.08	1.16	--
CPLX	0.70	0.85	1.00	1.15	1.30	1.65
Computer Attributes						
TIME	--	--	1.00	1.11	1.30	1.66
STOR	--	--	1.00	1.06	1.21	1.56
VIRT	--	0.87	1.00	1.15	1.30	--
TURN	--	0.87	1.00	1.07	1.15	--

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Cost Drivers	RATINGS					
	Very low	Low	Nominal	High	Very high	Extra high
Personnel Attributes						
ACAP	1.46	1.19	1.00	0.86	0.71	--
AEXP	1.29	1.13	1.00	0.91	0.82	--
PCAP	1.42	1.17	1.00	0.86	0.70	--
VEXP	1.21	1.10	1.00	0.90	--	--
LEXP	1.14	1.07	1.00	0.95	--	--
Project Attributes						
MODP	1.24	1.10	1.00	0.91	0.82	--
TOOL	1.24	1.10	1.00	0.91	0.83	--
SCED	1.23	1.08	1.00	1.04	1.10	--

Table 5: Multiplier values for effort calculations

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Intermediate COCOMO equations

$$E = a_i (KLOC)^{b_i} * EAF$$

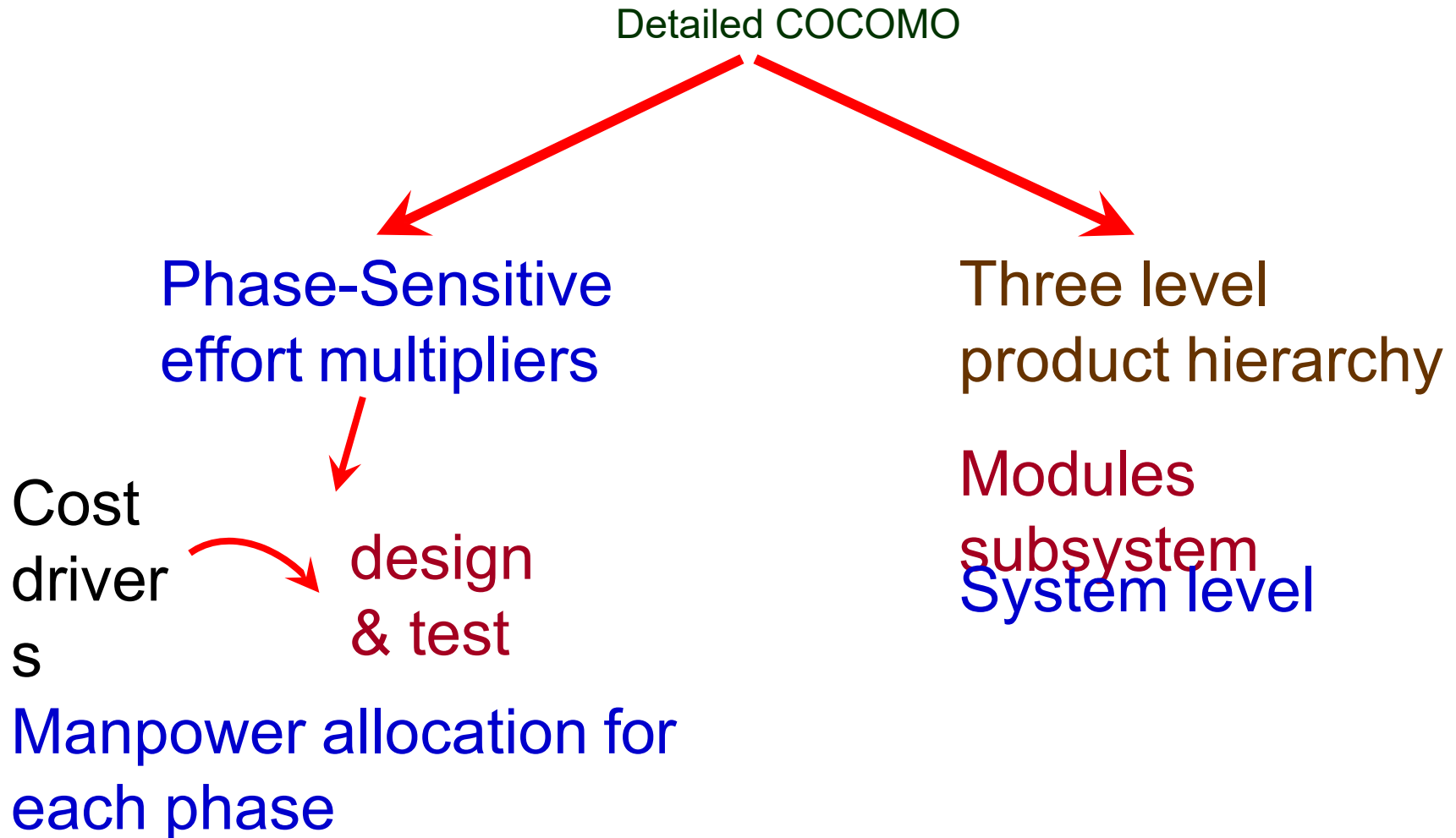
$$D = c_i (E)^{d_i}$$

Project	a_i	b_i	c_i	d_i
Organic	3.2	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	2.8	1.20	2.5	0.32

Table 6: Coefficients for intermediate COCOMO

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Detailed COCOMO Model



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Development Phase

Plan / Requirements

EFFORT : 6% to 8%

DEVELOPMENT TIME : 10% to

% depend on mode & size^{40%}

Software Project Planning

Design

Effort	:	16% to 18%
Time	:	19% to 38%

Programming

Effort	:	48% to 68%
Time	:	24% to 64%

Integration & Test

Effort	:	16% to 34%
Time	:	18% to 34%

Software Project Planning

Principle of the effort estimate

Size equivalent

As the software might be partly developed from software already existing (that is, re-usable code), a full development is not always required. In such cases, the parts of design document (DD%), code (C%) and integration (I%) to be modified are estimated. Then, an adjustment factor, A, is calculated by means of the following equation.

$$A = 0.4 \text{ DD} + 0.3 \text{ C} + 0.3 \text{ I}$$

The size equivalent is obtained by

$$S (\text{equivalent}) = (S \times A) / 100$$

$$E_p = \mu_p E D_p = \tau_p D$$

Software Project Planning

Lifecycle Phase Values of

$$\mu_p$$

Mode & Code Size	Plan & Requirements	System Design	Detailed Design	Module Code & Test	Integration & Test
Organic Small $S \approx 2$	0.06	0.16	0.26	0.42	0.16
Organic medium $S \approx 32$	0.06	0.16	0.24	0.38	0.22
Semidetached medium $S \approx 32$	0.07	0.17	0.25	0.33	0.25
Semidetached large $S \approx 128$	0.07	0.17	0.24	0.31	0.28
Embedded large $S \approx 128$	0.08	0.18	0.25	0.26	0.31
Embedded extra large $S \approx 320$	0.08	0.18	0.24	0.24	0.34

Table 7 : Effort and schedule fractions occurring in each phase of the lifecycle

Software Project Planning

Lifecycle Phase Values of

$$\tau_p$$

Mode & Code Size	Plan & Requirements	System Design	Detailed Design	Module Code & Test	Integration & Test
Organic Small $S \approx 2$	0.10	0.19	0.24	0.39	0.18
Organic medium $S \approx 32$	0.12	0.19	0.21	0.34	0.26
Semidetached medium $S \approx 32$	0.20	0.26	0.21	0.27	0.26
Semidetached large $S \approx 128$	0.22	0.27	0.19	0.25	0.29
Embedded large $S \approx 128$	0.36	0.36	0.18	0.18	0.28
Embedded extra large $S \approx 320$	0.40	0.38	0.16	0.16	0.30

Table 7 : Effort and schedule fractions occurring in each phase of the lifecycle

Software Project Planning

Distribution of software life cycle:

1. Requirement and product design
 - a) Plans and requirements
 - b) System design

1. Detailed Design
 - a) Detailed design

1. Code & Unit test
 - a) Module code & test

1. Integrate and Test
 - a) Integrate & Test

Software Project Planning

Example: 4.7

A new project with estimated 400 KLOC embedded system has to be developed. Project manager has a choice of hiring from two pools of developers: Very highly capable with very little experience in the programming language being used

Or

Developers of low quality but a lot of experience with the programming language. What is the impact of hiring all developers from one or the other pool ?

Software Project Planning

Solution

This is the case of embedded mode and model is intermediate COCOMO

Hence
$$E = a_i (KLOC)^{d_i}$$
$$= 2.8 (400)^{1.20} = 3712 \text{ PM}$$

Case I: Developers are very highly capable with very little experience in the programming being used.

$$EAF = 0.82 \times 1.14 = 0.9348$$

$$E = 3712 \times .9348 = 3470 \text{ PM}$$

$$D = 2.5 (3470)^{0.32} = 33.9 \text{ M}$$

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Case II: Developers are of low quality but lot of experience with the programming language being used.

$$EAF = 1.29 \times 0.95 = 1.22$$

$$E = 3712 \times 1.22 = 4528 \text{ PM}$$

$$D = 2.5 (4528)^{0.32} = 36.9 \text{ M}$$

Case II requires more effort and time. Hence, low quality developers with lot of programming language experience could not match with the performance of very highly capable developers with very little experience.

Software Project Planning

Example: 4.8

Consider a project to develop a full screen editor. The major components identified are:

- I. Screen edit
- II. Command Language Interpreter
- III. File Input & Output
- IV. Cursor Movement
- V. Screen Movement

The size of these are estimated to be 4k, 2k, 1k, 2k and 3k delivered source code lines. Use COCOMO to determine

1. Overall cost and schedule estimates (assume values for different cost drivers, with at least three of them being different from 1.0)
2. Cost & Schedule estimates for different

Software Project Planning

Solution

Size of five modules are:

Screen edit	= 4 KLOC
Command language interpreter	= 2 KLOC
File input and output	= 1 KLOC
Cursor movement Screen movement	= 2 KLOC
Total	= 3 KLOC
	= 12 KLOC

Software Project Planning

Let us assume that significant cost drivers are

- i. Required software reliability is high, i.e., 1.15
- ii. Product complexity is high, i.e., 1.15
- iii. Analyst capability is high, i.e., 0.86
- iv. Programming language experience is low, i.e., 1.07
- v. All other drivers are nominal

$$\text{EAF} = 1.15 \times 1.15 \times 0.86 \times 1.07 = 1.2169$$

Software Project Planning

(a) The initial effort estimate for the project is obtained from the following equation

$$E = a_i (\text{KLOC})^{b_i} \times \text{EAF}$$
$$= 3.2(12)^{1.05} \times 1.2169 = 52.91 \text{ PM}$$

Development time

$$D = C_i(E)^{d_i}$$
$$= 2.5(52.91)^{0.38} = 11.29 \text{ M}$$

(b) Using the following equations and referring Table 7, phase wise cost and schedule estimates can be calculated.

$$E_p = \mu_p E$$

$$D_p = \tau_p D$$

Software Project Planning

Since size is only 12 KLOC, it is an organic small model. Phase wise effort distribution is given below:

System Design	$= 0.16 \times 52.91 = 8.465 \text{ PM}$
Detailed Design	$= 0.26 \times 52.91 = 13.756 \text{ PM}$
Module Code & Test	$= 0.42 \times 52.91 = 22.222 \text{ PM}$
Integration & Test	$= 0.16 \times 52.91 = 8.465 \text{ Pm}$

Now Phase wise development time duration is

System Design	$= 0.19 \times 11.29 = 2.145 \text{ M}$
Detailed Design	$= 0.24 \times 11.29 = 2.709 \text{ M}$
Module Code & Test	$= 0.39 \times 11.29 = 4.403 \text{ M}$
Integration & Test	$= 0.18 \times 11.29 = 2.032 \text{ M}$

Software Project Planning

COCOMO-II

The following categories of applications/projects are identified by COCOMO-II and are shown in fig. 4 shown below:

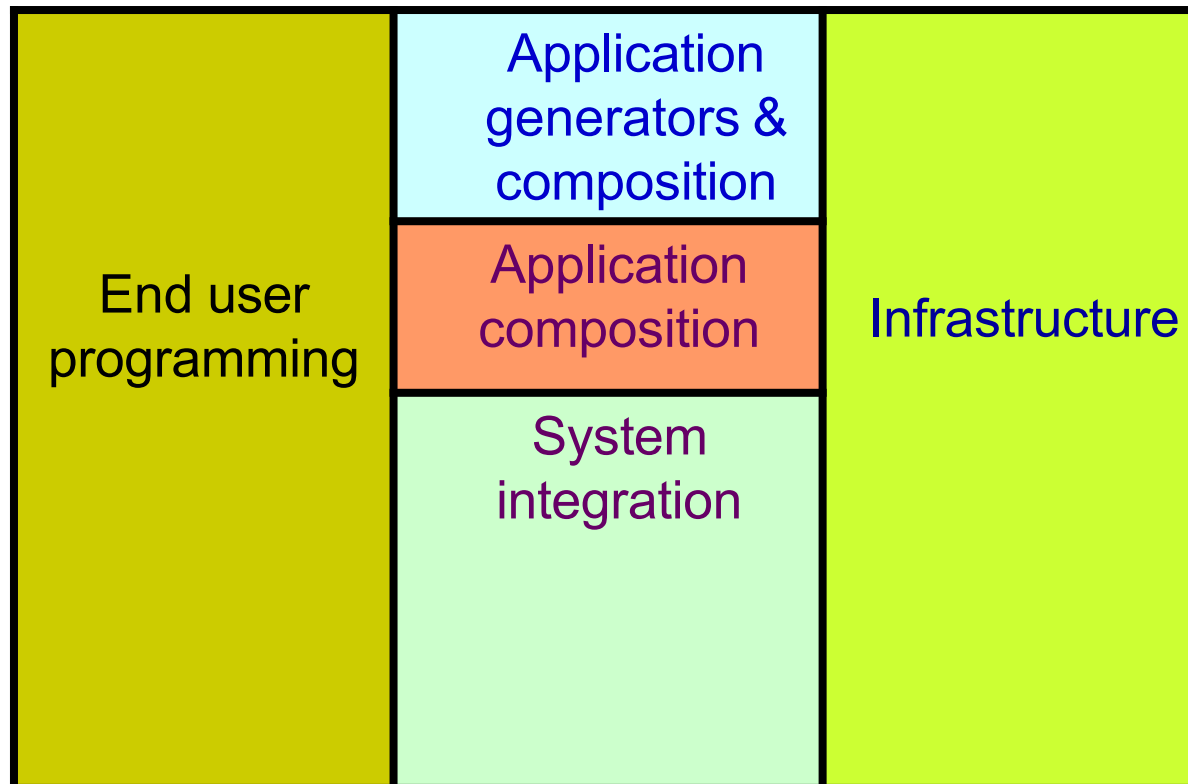


Fig. 4 : Categories of applications / projects

Software Project Planning

Stage No	Model Name	Application for the types of projects	Applications
Stage I	Application composition estimation model	Application composition	In addition to application composition type of projects, this model is also used for prototyping (if any) stage of application generators, infrastructure & system integration.
Stage II	Early design estimation model	Application generators, infrastructure & system integration	Used in early design stage of a project, when less is known about the project.
Stage III	Post architecture estimation model	Application generators, infrastructure & system integration	Used after the completion of the detailed architecture of the project.

Table 8: Stages of COCOMO-II

Software Project Planning

Application Composition Estimation Model

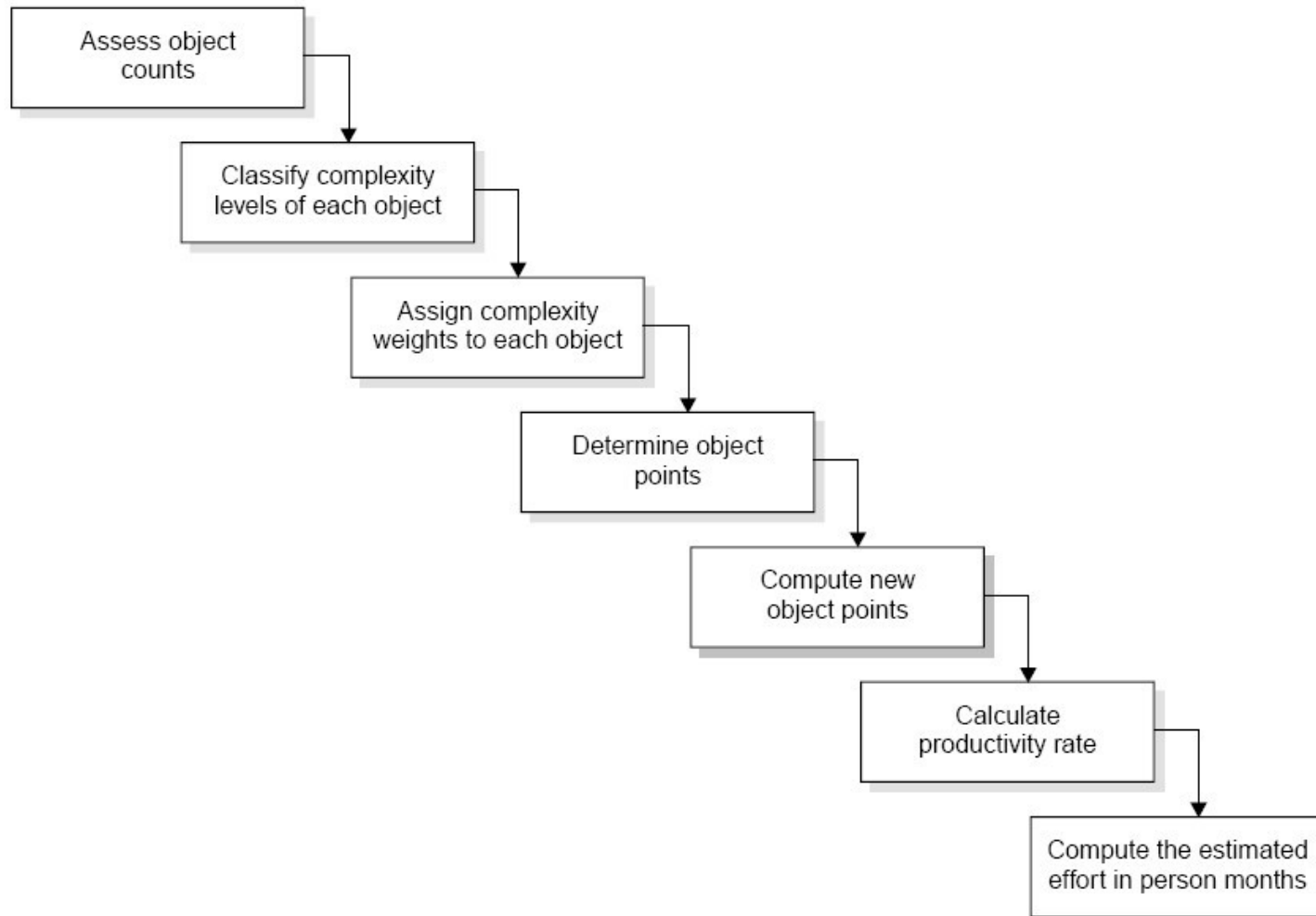


Fig.5: Steps for the estimation of effort in person months